

What is Data?

Data is a collection of a distinct small unit of information. It can be used in a variety of forms like text, numbers, media, bytes, etc. it can be stored in pieces of paper or electronic memory, etc.

Word 'Data' is originated from the word 'datum' that means 'single piece of information.' It is plural of the word datum.

In computing, Data is information that can be translated into a form for efficient movement and processing. Data is interchangeable.

What is Database?

*A **database** is an organized collection of data, so that it can be easily accessed and managed.*

You can organize data into tables, rows, columns, and index it to make it easier to find relevant information.

***Database handlers** create a database in such a way that only one set of software program provides access of data to all the users.*

*The **main purpose** of the database is to operate a large amount of information by storing, retrieving, and managing data.*

*There are many **dynamic websites** on the World Wide Web nowadays which are handled through databases. For example, a model that checks the availability of rooms in a hotel. It is an example of a dynamic website that uses a database.*

*There are many **databases available** like MySQL, Sybase, Oracle, MongoDB, Informix, PostgreSQL, SQL Server, etc.*

Modern databases are managed by the database management system (DBMS).

***SQL** or Structured Query Language is used to operate on the data stored in a database. SQL depends on relational algebra and tuple relational calculus.*

A cylindrical structure is used to display the image of a database.



Evolution of Databases

The database has completed more than 50 years of journey of its evolution from flat-file system to relational and objects relational systems. It has gone through several generations.

The Evolution

File-Based

1968 was the year when File-Based database were introduced. In file-based databases, data was maintained in a flat file. Though files have many advantages, there are several limitations.

One of the major advantages is that the file system has various access methods, e.g., sequential, indexed, and random.

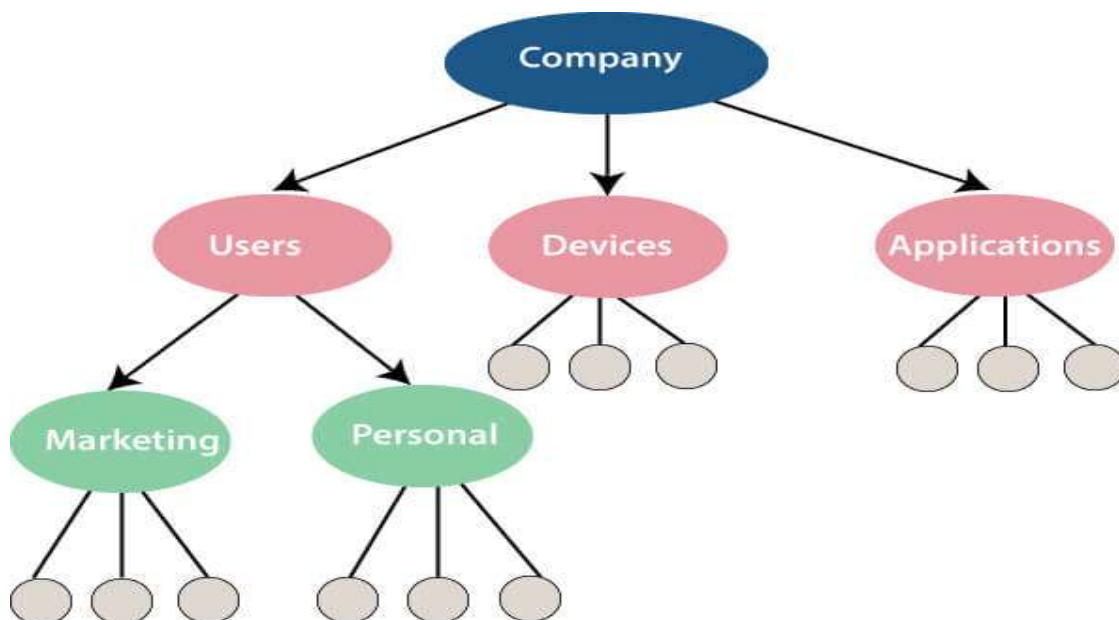
It requires extensive programming in a third-generation language such as COBOL, BASIC.

Hierarchical Data Model

1968-1980 was the era of the Hierarchical Database. Prominent hierarchical database model was IBM's first DBMS. It was called IMS (Information Management System).

In this model, files are related in a parent/child manner.

Below diagram represents Hierarchical Data Model. Small circle represents objects.



Like file system, this model also had some limitations like complex implementation, lack structural independence, can't easily handle a many-many relationship, etc.

Network data model

Charles Bachman developed the first DBMS at Honeywell called Integrated Data Store (IDS). It was developed in the early 1960s, but it was standardized in 1971 by the CODASYL group (Conference on Data Systems Languages).

In this model, files are related as owners and members, like to the common network model.

Network data model identified the following components:

- *Network schema (Database organization)*
- *Sub-schema (views of database per user)*
- *Data management language (procedural)*

This model also had some limitations like system complexity and difficult to design and maintain.

Relational Database

1970 - Present: *It is the era of Relational Database and Database Management. In 1970, the relational model was proposed by E.F. Codd.*

Relational database model has two main terminologies called instance and schema.

The instance is a table with rows or columns

Schema specifies the structure like name of the relation, type of each column and name.

This model uses some mathematical concept like set theory and predicate logic.

The first internet database application had been created in 1995.

During the era of the relational database, many more models had introduced like object-oriented model, object-relational model, etc.

Cloud database

Cloud database facilitates you to store, manage, and retrieve their structured, unstructured data via a cloud platform. This data is accessible over the Internet. Cloud databases are also called a database as service (DBaaS) because they are offered as a managed service.

Some best cloud options are:

- *AWS (Amazon Web Services)*
- *Snowflake Computing*
- *Oracle Database Cloud Services*
- *Microsoft SQL server*
- *Google cloud spanner*

Advantages of cloud database

Lower costs

Generally, company provider does not have to invest in databases. It can maintain and support one or more data centers.

Automated

Cloud databases are enriched with a variety of automated processes such as recovery, failover, and auto-scaling.

Increased accessibility

You can access your cloud-based database from any location, anytime. All you need is just an internet connection.

NoSQL Database

A NoSQL database is an approach to design such databases that can accommodate a wide variety of data models. NoSQL stands for "not only SQL." It is an alternative to traditional relational databases in which data is placed in tables, and data schema is perfectly designed before the database is built.

NoSQL databases are useful for a large set of distributed data.

Some examples of NoSQL database system with their category are:

- *MongoDB, CouchDB, Cloudant (Document-based)*
- *Memcached, Redis, Coherence (key-value store)*
- *HBase, Big Table, Accumulo (Tabular)*

Advantage of NoSQL

High Scalability

NoSQL can handle an extensive amount of data because of scalability. If the data grows, NoSQL database scale it to handle that data in an efficient manner.

High Availability

NoSQL supports auto replication. Auto replication makes it highly available because, in case of any failure, data replicates itself to the previous consistent state.

Disadvantage of NoSQL

Open source

NoSQL is an open-source database, so there is no reliable standard for NoSQL yet.

Management challenge

Data management in NoSQL is much more complicated than relational databases. It is very challenging to install and even more hectic to manage daily.

GUI is not available

GUI tools for NoSQL database are not easily available in the market.

Backup

Backup is a great weak point for NoSQL databases. Some databases, like MongoDB, have no powerful approaches for data backup.

The Object-Oriented Databases

The object-oriented databases contain data in the form of object and classes. Objects are the real-world entity, and types are the collection of objects. An object-oriented database is a combination of relational model features with objects oriented principles. It is an alternative implementation to that of the relational model.

Object-oriented databases hold the rules of object-oriented programming. An object-oriented database management system is a hybrid application.

The object-oriented database model contains the following properties.

Object-oriented programming properties

- *Objects*
- *Classes*
- *Inheritance*

- *Polymorphism*
- *Encapsulation*

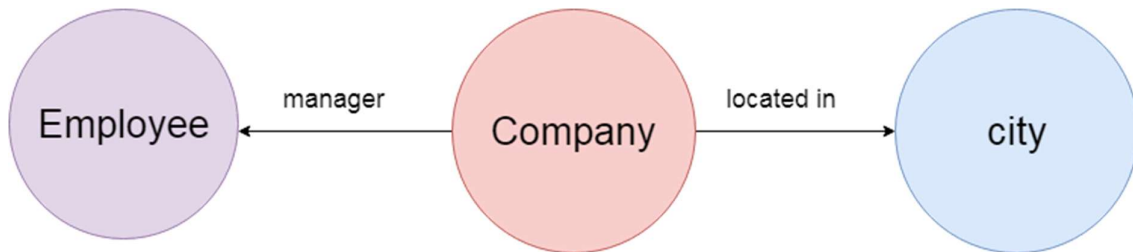
Relational database properties

- *Atomicity*
- *Consistency*
- *Integrity*
- *Durability*
- *Concurrency*
- *Query processing*

Graph Databases

A graph database is a NoSQL database. It is a graphical representation of data. It contains nodes and edges. A node represents an entity, and each edge represents a relationship between two edges. Every node in a graph database represents a unique identifier.

Graph databases are beneficial for searching the relationship between data because they highlight the relationship between relevant data.



Graph databases are very useful when the database contains a complex relationship and dynamic schema.

*It is mostly used in **supply chain management**, identifying the source of **IP telephony**.*

DBMS (Data Base Management System)

Database management System is software which is used to store and retrieve the database. For example, Oracle, MySQL, etc.; these are some popular DBMS tools.

- *DBMS provides the interface to perform the various operations like creation, deletion, modification, etc.*
- *DBMS allows the user to create their databases as per their requirement.*

- DBMS accepts the request from the application and provides specific data through the operating system.
- DBMS contains the group of programs which acts according to the user instruction.
- It provides security to the database.

Advantage of DBMS

Controls redundancy

It stores all the data in a single database file, so it can control data redundancy.

Data sharing

An authorized user can share the data among multiple users.

Backup

It provides Backup and recovery subsystem. This recovery system creates automatic data from system failure and restores data if required.

Multiple user interfaces

It provides a different type of user interfaces like GUI, application interfaces.

Disadvantage of DBMS

Size

It occupies large disk space and large memory to run efficiently.

Cost

DBMS requires a high-speed data processor and larger memory to run DBMS software, so it is costly.

Complexity

DBMS creates additional complexity and requirements.

RDBMS (Relational Database Management System)

The word RDBMS is termed as 'Relational Database Management System.' It is represented as a table that contains rows and column.

RDBMS is based on the Relational model; it was introduced by E. F. Codd.

A relational database contains the following components:

- *Table*
- *Record/ Tuple*
- *Field/Column name /Attribute*
- *Instance*
- *Schema*
- *Keys*

An RDBMS is a tabular DBMS that maintains the security, integrity, accuracy, and consistency of the data.

Database defined

A database is an organized collection of structured information, or data, typically stored electronically in a computer system. A database is usually controlled by a database management system (DBMS). Together, the data and the DBMS, along with the applications that are associated with them, are referred to as a database system, often shortened to just database.

Data within the most common types of databases in operation today is typically modeled in rows and columns in a series of tables to make processing and data querying efficient. The data can then be easily accessed, managed, modified, updated, controlled, and organized. Most databases use structured query language (SQL) for writing and querying data.

What is Structured Query Language (SQL)?

SQL is a programming language used by nearly all relational databases to query, manipulate, and define data, and to provide access control. SQL was first developed at IBM in the 1970s with Oracle as a major contributor, which led to implementation of the SQL ANSI standard, SQL has spurred many extensions from companies such as IBM, Oracle, and Microsoft. Although SQL is still widely used today, new programming languages are beginning to appear.

Evolution of the database

Databases have evolved dramatically since their inception in the early 1960s. Navigational databases such as the hierarchical database (which relied on a tree-like model and allowed only a one-to-many relationship), and the network database (a more flexible model that allowed multiple relationships), were the original systems used to store and manipulate data. Although simple, these early systems were inflexible. In the 1980s, relational databases became popular, followed by object-oriented databases in the 1990s. More recently, NoSQL databases came about as a response to the growth of the internet and the need for faster speed and processing of unstructured data. Today, cloud databases and self-driving databases are breaking new ground when it comes to how data is collected, stored, managed, and utilized.

What's the difference between a database and a spreadsheet?

Databases and spreadsheets (such as Microsoft Excel) are both convenient ways to store information. The primary differences between the two are:

- How the data is stored and manipulated*
- Who can access the data*
- How much data can be stored*

Spreadsheets were originally designed for one user, and their characteristics reflect that. They're great for a single user or small number of users who don't need to do a lot of incredibly complicated data manipulation. Databases, on the other hand, are designed to hold much larger collections of organized information—massive amounts, sometimes. Databases allow multiple users at the same time to quickly and securely access and query the data using highly complex logic and language.

Types of databases

There are many different types of databases. The best database for a specific organization depends on how the organization intends to use the data.

Relational databases

- *Relational databases became dominant in the 1980s. Items in a relational database are organized as a set of tables with columns and rows. Relational database technology provides the most efficient and flexible way to access structured information.*

Object-oriented databases

- *Information in an object-oriented database is represented in the form of objects, as in object-oriented programming.*

Distributed databases

- *A distributed database consists of two or more files located in different sites. The database may be stored on multiple computers, located in the same physical location, or scattered over different networks.*

Data warehouses

- *A central repository for data, a data warehouse is a type of database specifically designed for fast query and analysis.*

NoSQL databases

- *A NoSQL, or nonrelational database, allows unstructured and semistructured data to be stored and manipulated (in contrast to a relational database, which defines how all data inserted into the database must be composed). NoSQL databases grew popular as web applications became more common and more complex.*

Graph databases

- *A graph database stores data in terms of entities and the relationships between entities.*

- **OLTP databases.** An OLTP database is a speedy, analytic database designed for large numbers of transactions performed by multiple users.

These are only a few of the several dozen types of databases in use today. Other, less common databases are tailored to very specific scientific, financial, or other functions. In addition to the different database types, changes in technology development approaches and dramatic advances such as the cloud and automation are propelling databases in entirely new directions. Some of the latest databases include

Open source databases

- An open source database system is one whose source code is open source; such databases could be SQL or NoSQL databases.

Cloud databases

- A cloud database is a collection of data, either structured or unstructured, that resides on a private, public, or hybrid cloud computing platform. There are two types of cloud database models: traditional and database as a service (DBaaS). With DBaaS, administrative tasks and maintenance are performed by a service provider.

Multimodel database

- Multimodel databases combine different types of database models into a single, integrated back end. This means they can accommodate various data types.

Document/JSON database

- Designed for storing, retrieving, and managing document-oriented information, document databases are a modern way to store data in JSON format rather than rows and columns.

Self-driving databases

- The newest and most groundbreaking type of database, self-driving databases (also known as autonomous databases) are cloud-based and use machine learning to automate database tuning, security, backups, updates, and other routine management tasks traditionally performed by database administrators.

What is database software?

Database software is used to create, edit, and maintain database files and records, enabling easier file and record creation, data entry, data editing, updating, and reporting. The software also handles data storage, backup and reporting, multi-access control, and security. Strong database security is especially important today, as data theft becomes more frequent.

Database software is sometimes also referred to as a “database management system” (DBMS).

Database software makes data management simpler by enabling users to store data in a structured form and then access it. It typically has a graphical interface to help create and manage the data and, in some cases, users can construct their own databases by using database software.

What is a database management system (DBMS)?

A database typically requires a comprehensive database software program known as a database management system (DBMS). A DBMS serves as an interface between the database and its end users or programs, allowing users to retrieve, update, and manage how the information is organized and optimized. A DBMS also facilitates oversight and control of databases, enabling a variety of administrative operations such as performance monitoring, tuning, and backup and recovery.

Some examples of popular database software or DBMSs include MySQL, Microsoft Access, Microsoft SQL Server, FileMaker Pro, Oracle Database, and dBASE.

What is a MySQL database?

MySQL is an open source relational database management system based on SQL. It was designed and optimized for web applications and can run on any platform. As new and different requirements emerged with the internet, MySQL became the platform of choice for web developers and web-based applications. Because it's designed to process millions of queries and thousands of transactions, MySQL is a popular choice for ecommerce businesses that need to manage multiple money transfers. On-demand flexibility is the primary feature of MySQL.

MySQL is the DBMS behind some of the top websites and web-based applications in the world, including Airbnb, Uber, LinkedIn, Facebook, Twitter, and YouTube.

Using databases to improve business performance and decision-making

With massive data collection from the Internet of Things transforming life and industry across the globe, businesses today have access to more data than ever before. Forward-thinking organizations can now use databases to go beyond basic data storage and transactions to analyze vast quantities of data from multiple systems. Using database and other computing and business intelligence tools, organizations can now leverage the data they collect to run more efficiently, enable better decision-making, and become more agile and scalable. Optimizing access and throughput to data is critical to businesses today because there is more data volume to track. It's critical to have a platform that can deliver the performance, scale, and agility that businesses need as they grow over time.

The self-driving database is poised to provide a significant boost to these capabilities. Because self-driving databases automate expensive, time-consuming manual processes, they free up

business users to become more proactive with their data. By having direct control over the ability to create and use databases, users gain control and autonomy while still maintaining important security standards.

Database challenges

Today's large enterprise databases often support very complex queries and are expected to deliver nearly instant responses to those queries. As a result, database administrators are constantly called upon to employ a wide variety of methods to help improve performance. Some common challenges that they face include:

- **Absorbing significant increases in data volume.** The explosion of data coming in from sensors, connected machines, and dozens of other sources keeps database administrators scrambling to manage and organize their companies' data efficiently.
- **Ensuring data security.** Data breaches are happening everywhere these days, and hackers are getting more inventive. It's more important than ever to ensure that data is secure but also easily accessible to users.
- **Keeping up with demand.** In today's fast-moving business environment, companies need real-time access to their data to support timely decision-making and to take advantage of new opportunities.
- **Managing and maintaining the database and infrastructure.** Database administrators must continually watch the database for problems and perform preventative maintenance, as well as apply software upgrades and patches. As databases become more complex and data volumes grow, companies are faced with the expense of hiring additional talent to monitor and tune their databases.
- **Removing limits on scalability.** A business needs to grow if it's going to survive, and its data management must grow along with it. But it's very difficult for database administrators to predict how much capacity the company will need, particularly with on-premises databases.
- **Ensuring data residency, data sovereignty, or latency requirements.** Some organizations have use cases that are better suited to run on-premises. In those cases, engineered systems that are pre-configured and pre-optimized for running the database are ideal. Customers achieve higher availability, greater performance and up to 40% lower cost with Oracle Exadata, according to [Wikibon's recent analysis \(PDF\)](#).

Addressing all of these challenges can be time-consuming and can prevent database administrators from performing more strategic functions.

How autonomous technology is improving database management

Self-driving databases are the wave of the future—and offer an intriguing possibility for organizations that want to use the best available database technology without the headaches of running and operating that technology.

Self-driving databases use cloud-based technology and machine learning to automate many of the routine tasks required to manage databases, such as tuning, security, backups, updates, and other routine management tasks. With these tedious tasks automated, database administrators are freed up to do more strategic work. The self-driving, self-securing, and self-repairing capabilities of self-driving databases are poised to revolutionize how companies manage and secure their data, enabling performance advantages, lower costs, and improved security.

Relational Calculus

There is an alternate way of formulating queries known as Relational Calculus. Relational calculus is a non-procedural query language. In the non-procedural query language, the user is concerned with the details of how to obtain the end results. The relational calculus tells what to do but never explains how to do. Most commercial relational languages are based on aspects of relational calculus including SQL-QBE and QUEL.

Why it is called Relational Calculus?

It is based on Predicate calculus, a name derived from branch of symbolic language. A predicate is a truth-valued function with arguments. On substituting values for the arguments, the function result in an expression called a proposition. It can be either true or false. It is a tailored version of a subset of the Predicate Calculus to communicate with the relational database.

Many of the calculus expressions involves the use of Quantifiers. There are two types of quantifiers:

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Exception Handling in Java - Javatpoint

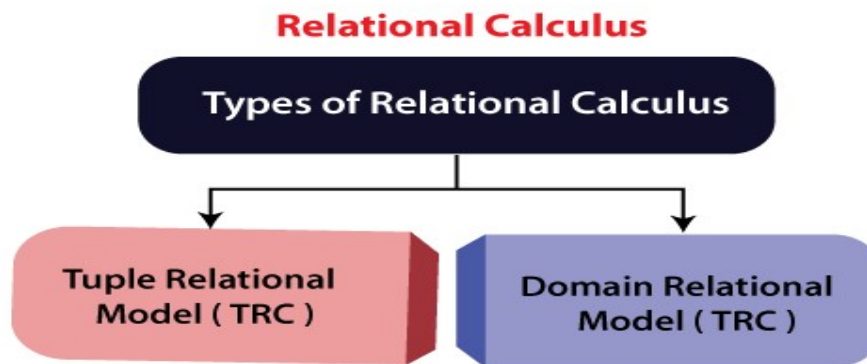
- ***Universal Quantifiers:*** *The universal quantifier denoted by $\forall$ is read as for all which means that in a given set of tuples exactly all tuples satisfy a given condition.*
- ***Existential Quantifiers:*** *The existential quantifier denoted by $\exists$ is read as for all which means that in a given set of tuples there is at least one occurrences whose value satisfy a given condition.*

Before using the concept of quantifiers in formulas, we need to know the concept of Free and Bound Variables.

A tuple variable t is bound if it is quantified which means that if it appears in any occurrences a variable that is not bound is said to be free.

Free and bound variables may be compared with global and local variable of programming languages.

Types of Relational calculus:



1. Tuple Relational Calculus (TRC)

It is a non-procedural query language which is based on finding a number of tuple variables also known as range variable for which predicate holds true. It describes the desired information without giving a specific procedure for obtaining that information. The tuple relational calculus is specified to select the tuples in a relation. In TRC, filtering variable uses the tuples of a relation. The result of the relation can have one or more tuples.

Notation:

A Query in the tuple relational calculus is expressed as following notation

$$1. \{T \mid \mathcal{P}(T)\} \text{ or } \{T \mid \text{Condition}(T)\}$$

Where

T is the resulting tuples

$\mathcal{P}(T)$ is the condition used to fetch T .

For example:

$$1. \{T.name \mid Author(T) \text{ AND } T.article = 'database'\}$$

Output: This query selects the tuples from the AUTHOR relation. It returns a tuple with 'name' from Author who has written an article on 'database'.

TRC (tuple relation calculus) can be quantified. In TRC, we can use Existential ($\exists$) and Universal Quantifiers ($\forall$).

For example:

1. $\{R \mid \exists T \in \text{Authors}(T.\text{article} = \text{'database'} \wedge R.\text{name} = T.\text{name})\}$

Output: This query will yield the same result as the previous one.

2. Domain Relational Calculus (DRC)

The second form of relation is known as Domain relational calculus. In domain relational calculus, filtering variable uses the domain of attributes. Domain relational calculus uses the same operators as tuple calculus. It uses logical connectives $\wedge$ (and), $\vee$ (or) and $\neg$ (not). It uses Existential ($\exists$) and Universal Quantifiers ($\forall$) to bind the variable. The QBE or Query by example is a query language related to domain relational calculus.

Notation:

1. $\{a_1, a_2, a_3, \dots, a_n \mid P(a_1, a_2, a_3, \dots, a_n)\}$

Where

a_1, a_2 are attributes
 P stands for formula built by inner attributes

For example:

1. $\{ \langle \text{article, page, subject} \rangle \mid \in \text{javatpoint} \wedge \text{subject} = \text{'database'} \}$

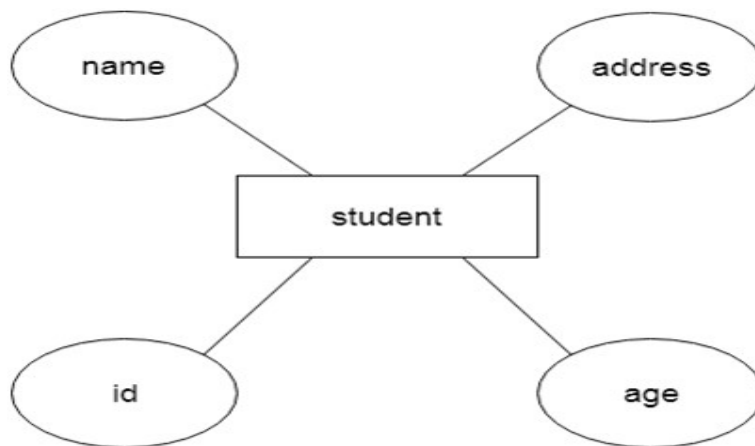
Output: This query will yield the article, page, and subject from the relational javatpoint, where the subject is a database.

ER (Entity Relationship) Diagram in DBMS

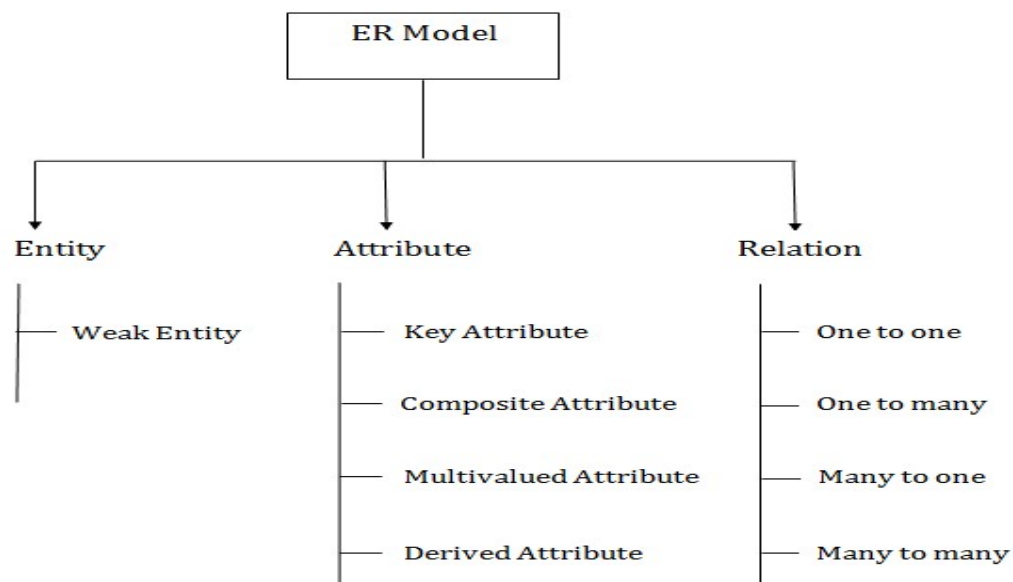
- ER model stands for an Entity-Relationship model. It is a high-level data model. This model is used to define the data elements and relationship for a specified system.

- It develops a conceptual design for the database. It also develops a very simple and easy to design view of data.
- In ER modeling, the database structure is portrayed as a diagram called an entity-relationship diagram.

For example, Suppose we design a school database. In this database, the student will be an entity with attributes like address, name, id, age, etc. The address can be another entity with attributes like city, street name, pin code, etc and there will be a relationship between them.



Component of ER Diagram



1. Entity:

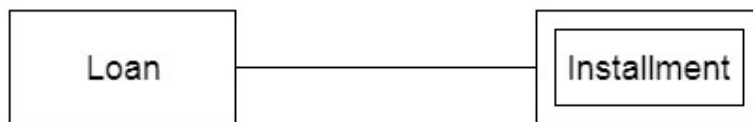
An entity may be any object, class, person or place. In the ER diagram, an entity can be represented as rectangles.

Consider an organization as an example- manager, product, employee, department etc. can be taken as an entity.



a. Weak Entity

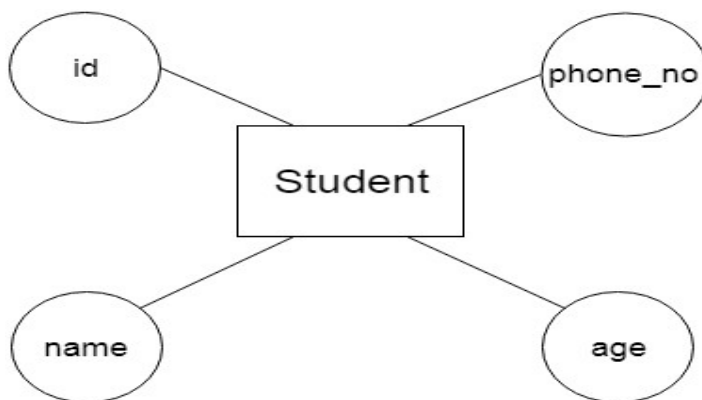
An entity that depends on another entity called a weak entity. The weak entity doesn't contain any key attribute of its own. The weak entity is represented by a double rectangle.



2. Attribute

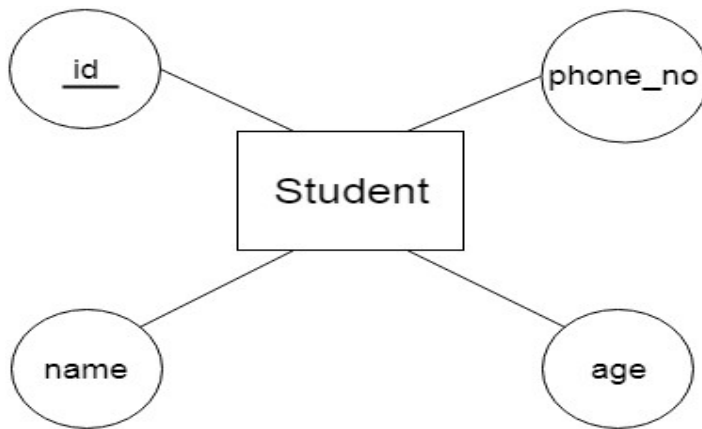
The attribute is used to describe the property of an entity. Eclipse is used to represent an attribute.

For example, id, age, contact number, name, etc. can be attributes of a student.



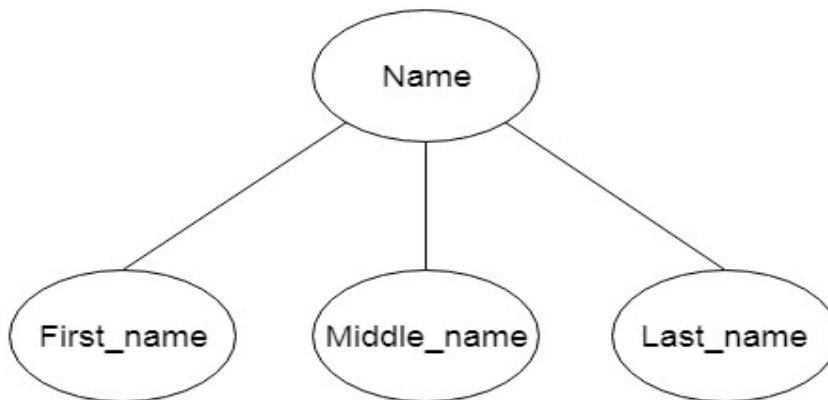
a. Key Attribute

The key attribute is used to represent the main characteristics of an entity. It represents a primary key. The key attribute is represented by an ellipse with the text underlined.



b. Composite Attribute

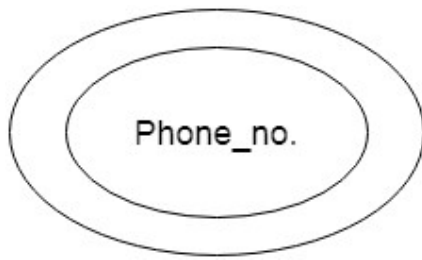
An attribute that composed of many other attributes is known as a composite attribute. The composite attribute is represented by an ellipse, and those ellipses are connected with an ellipse.



c. Multivalued Attribute

An attribute can have more than one value. These attributes are known as a multivalued attribute. The double oval is used to represent multivalued attribute.

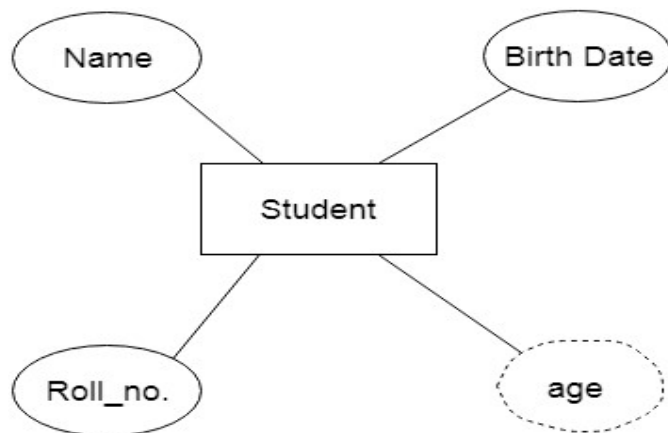
For example, a student can have more than one phone number.



d. Derived Attribute

An attribute that can be derived from other attribute is known as a derived attribute. It can be represented by a dashed ellipse.

For example, A person's age changes over time and can be derived from another attribute like Date of birth.



3. Relationship

A relationship is used to describe the relation between entities. Diamond or rhombus is used to represent the relationship.



Types of relationship are as follows:

a. One-to-One Relationship

When only one instance of an entity is associated with the relationship, then it is known as one to one relationship.

For example, *A female can marry to one male, and a male can marry to one female.*



b. One-to-many relationship

When only one instance of the entity on the left, and more than one instance of an entity on the right associates with the relationship then this is known as a one-to-many relationship.

For example, *Scientist can invent many inventions, but the invention is done by the only specific scientist.*



c. Many-to-one relationship

When more than one instance of the entity on the left, and only one instance of an entity on the right associates with the relationship then it is known as a many-to-one relationship.

For example, *Student enrolls for only one course, but a course can have many students.*



d. Many-to-many relationship

When more than one instance of the entity on the left, and more than one instance of an entity on the right associates with the relationship then it is known as a many-to-many relationship.

For example, Employee can assign by many projects and project can have many employees.



SQL

- *SQL stands for Structured Query Language. It is used for storing and managing data in relational database management system (RDBMS).*
- *It is a standard language for Relational Database System. It enables a user to create, read, update and delete relational databases and tables.*
- *All the RDBMS like MySQL, Informix, Oracle, MS Access and SQL Server use SQL as their standard database language.*
- *SQL allows users to query the database in a number of ways, using English-like statements.*

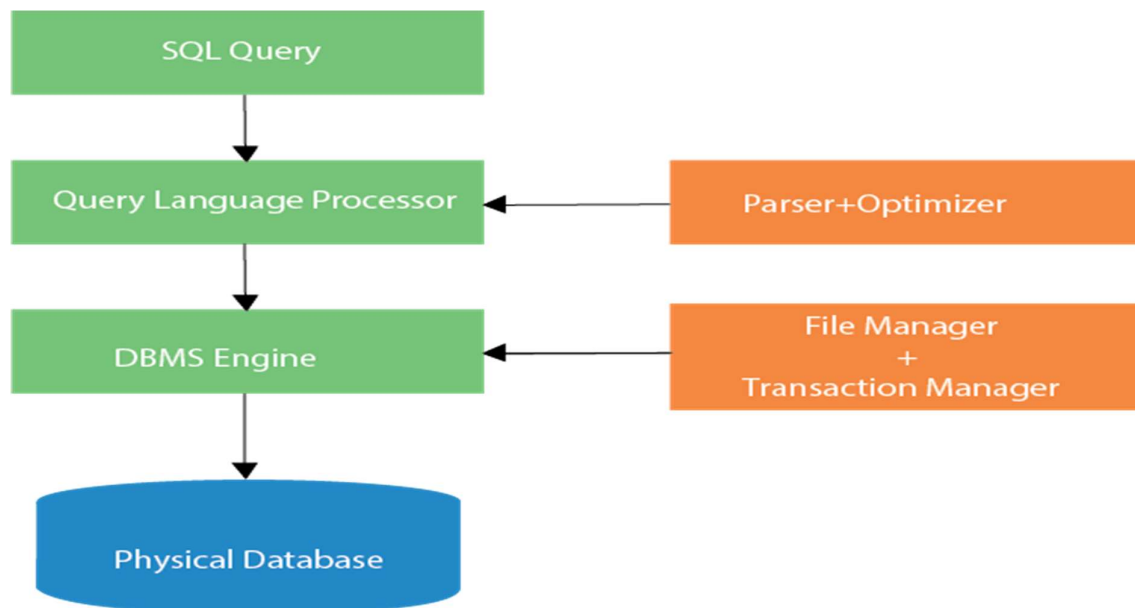
Rules:

SQL follows the following rules:

- *Structure query language is not case sensitive. Generally, keywords of SQL are written in uppercase.*
- *Statements of SQL are dependent on text lines. We can use a single SQL statement on one or multiple text line.*
- *Using the SQL statements, you can perform most of the actions in a database.*
- *SQL depends on tuple relational calculus and relational algebra.*

SQL process:

- *When an SQL command is executing for any RDBMS, then the system figure out the best way to carry out the request and the SQL engine determines that how to interpret the task,*
- *In the process, various components are included. These components can be optimization Engine, Query engine, Query dispatcher, classic, etc.*
- *All the non-SQL queries are handled by the classic query engine, but SQL query engine won't handle logical files.*



Characteristics of SQL

- *SQL is easy to learn.*
- *SQL is used to access data from relational database management systems.*
- *SQL can execute queries against the database.*
- *SQL is used to describe the data.*
- *SQL is used to define the data in the database and manipulate it when needed.*
- *SQL is used to create and drop the database and table.*
- *SQL is used to create a view, stored procedure, function in a database.*
- *SQL allows users to set permissions on tables, procedures, and views.*

Advantages of SQL

There are the following advantages of SQL:

High speed

Using the SQL queries, the user can quickly and efficiently retrieve a large amount of records from a database.

No coding needed

In the standard SQL, it is very easy to manage the database system. It doesn't require a substantial amount of code to manage the database system.

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Hello Java Program for Beginners

Well defined standards

Long established are used by the SQL databases that are being used by ISO and ANSI.

Portability

SQL can be used in laptop, PCs, server and even some mobile phones.

Interactive language

SQL is a domain language used to communicate with the database. It is also used to receive answers to the complex questions in seconds.

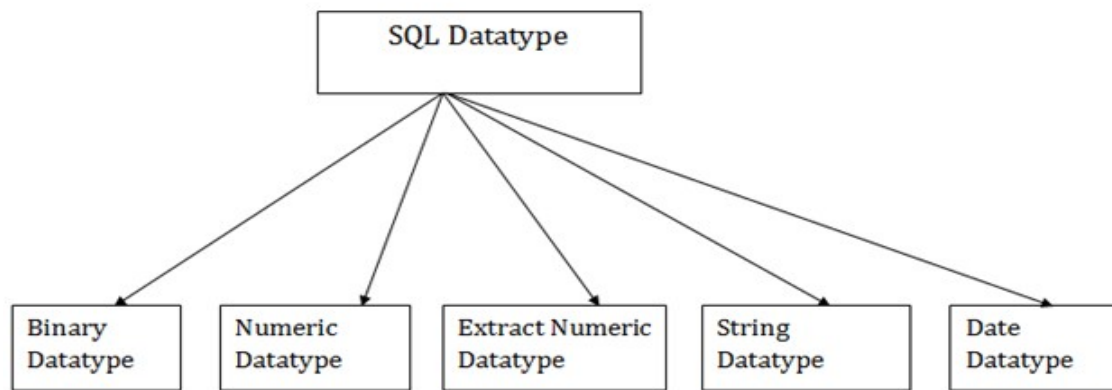
Multiple data view

Using the SQL language, the users can make different views of the database structure.

SQL Datatype

- *SQL Datatype is used to define the values that a column can contain.*
- *Every column is required to have a name and data type in the database table.*

Datatype of SQL:



1. Binary Datatypes

There are Three types of binary Datatypes which are given below:

<i>Data Type</i>	<i>Description</i>
<i>binary</i>	<i>It has a maximum length of 8000 bytes. It contains fixed-length binary data.</i>
<i>varbinary</i>	<i>It has a maximum length of 8000 bytes. It contains variable-length binary data.</i>
<i>image</i>	<i>It has a maximum length of 2,147,483,647 bytes. It contains variable-length binary data.</i>

2. Approximate Numeric Datatype :

The subtypes are given below:

<i>Data type</i>	<i>From</i>	<i>To</i>	<i>Description</i>
<i>float</i>	<i>-1.79E + 308</i>	<i>1.79E + 308</i>	<i>It is used to specify a floating-point value e.g. 6.2, 2.9 e</i>
<i>real</i>	<i>-3.40e + 38</i>	<i>3.40E + 38</i>	<i>It specifies a single precision floating point number</i>

3. Exact Numeric Datatype

The subtypes are given below:

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Data type	Description
<i>int</i>	<i>It is used to specify an integer value.</i>
<i>smallint</i>	<i>It is used to specify small integer value.</i>
<i>bit</i>	<i>It has the number of bits to store.</i>
<i>decimal</i>	<i>It specifies a numeric value that can have a decimal number.</i>
<i>numeric</i>	<i>It is used to specify a numeric value.</i>

4. Character String Datatype

The subtypes are given below:

Data type	Description
<i>char</i>	<i>It has a maximum length of 8000 characters. It contains Fixed-length non-unicode characters.</i>
<i>varchar</i>	<i>It has a maximum length of 8000 characters. It contains variable-length non-unicode characters.</i>
<i>text</i>	<i>It has a maximum length of 2,147,483,647 characters. It contains variable-length non-unicode c.</i>

5. Date and time Datatypes

The subtypes are given below:

Datatype	Description
<i>date</i>	<i>It is used to store the year, month, and days value.</i>

<i>time</i>	<i>It is used to store the hour, minute, and second values.</i>
<i>timestamp</i>	<i>It stores the year, month, day, hour, minute, and the second value.</i>